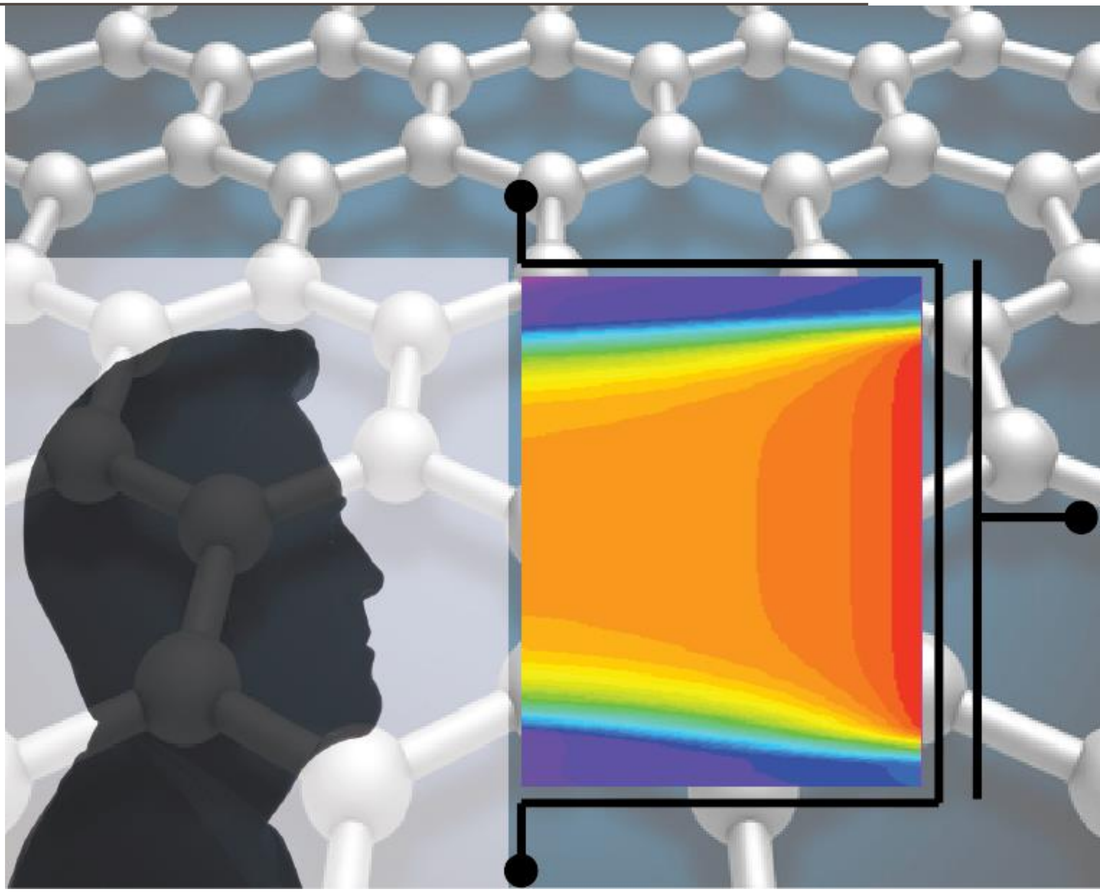
ADVANCED SOLID-STATE DEVICES

Week-5, Lecture-2



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6^{th} February, 2025

PN Junction

Reverse Bias

Junction Capacitance: One-sided junction

Problem: Compute the junction capacitance of a one-sided p^+n junction.

$$C_j = \sqrt{\frac{q\varepsilon_s N_a N_d}{2(N_a + N_d)(V_{bi} + V_R)}}$$

$$\frac{1}{{C_j}^2} = \frac{2(V_{bi} + V_R)}{q\varepsilon_s N_d}$$

Solution:

$$N_a \gg N_d$$

$$C_j \approx \sqrt{\frac{q\varepsilon_s N_a N_d}{2N_a(V_{bi} + V_R)}} = \sqrt{\frac{q\varepsilon_s N_d}{2(V_{bi} + V_R)}}$$

- By measuring capacitance at different voltage and plotting, we can determine:
  - ➢ Built in potential: X-intercept
  - ➢ Doping: slope

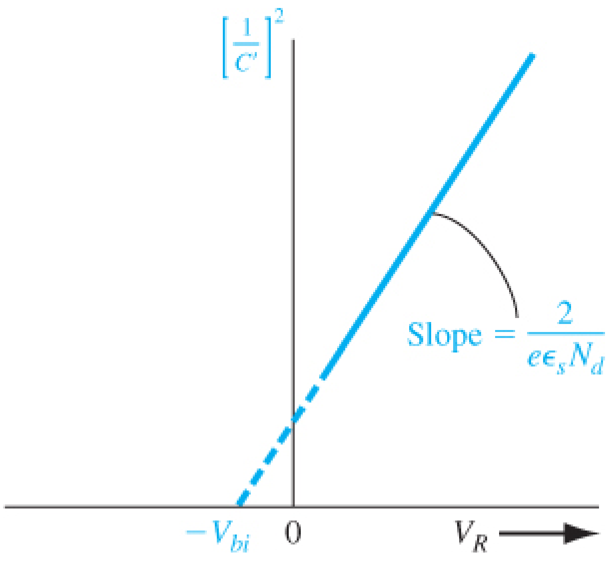

Figure 7.11 | $(1/C')^2$ versus V_R of a uniformly doped pn junction.

PN Junction

Current-Voltage Characteristics

Current-Voltage Relationship: Ideal Characteristics

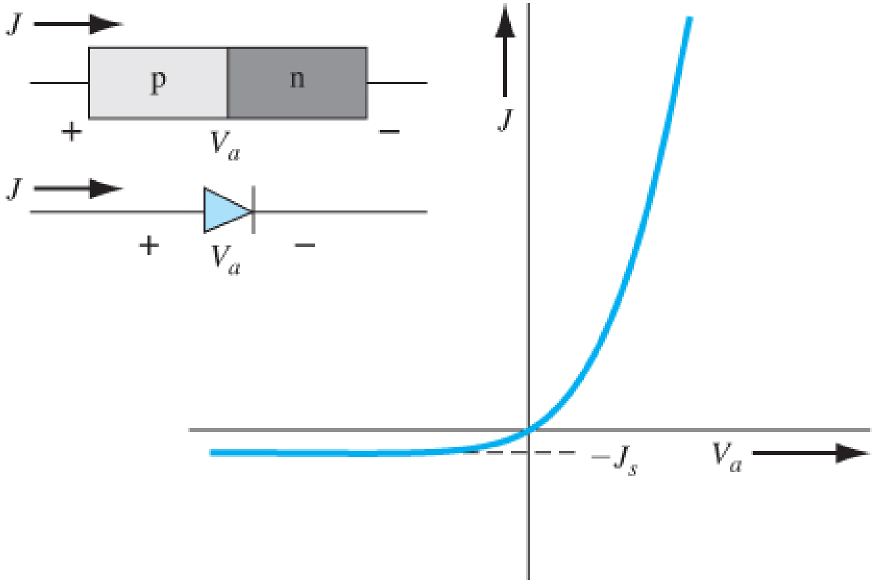

Figure 8.8 | Ideal *I–V* characteristic of a pn junction diode.

Two modes:

1. Forward-biased: exponential
2. Reverse-biased: constant (reverse saturation current J_S

Why does a diode exhibit this behaviour?

- Start with an ideal case
- Then make it more complicated (realistic)

Ideal (because following simplifying assumptions are used):

1. Depletion layer has abrupt boundaries
2. FD statistics can be approximated by MB statistics
3. Low level injection and complete ionization
4. Individual electron and hole currents are constant throughout the depletion region and continuous for the device

Notations

At equilibrium

On p-side: minority carrier electron: n_{p0} and majority carrier hole: p_{p0}

On n-side: minority carrier hole: p_{n0} and majority carrier electron: n_{n0}

When bias is applied:

On p-side: minority carrier electron: n_p and majority carrier hole: p_p

On n-side: minority carrier hole: p_n and majority carrier electron: n_n

Ideal Characteristics: Minority Carrier at Equilibrium

- Initially (at equilibrium), potential barrier is:

$$V_{bi} = V_t ln\left(\frac{N_a N_d}{n_i^2}\right) \Rightarrow \frac{n_i^2}{N_a N_d} = exp\left(-\frac{qV_{bi}}{kT}\right)$$

Assuming complete ionization:

$$\frac{n_i^2}{p_{p0} n_{n0}} = exp\left(-\frac{qV_{bi}}{kT}\right)$$

Minority carrier electrons on the p-side:

$$n_{p0} = \frac{n_i^2}{p_{p0}}$$

$n_{p0} = n_{n0} exp\left(-\frac{qV_{bi}}{kT}\right)$ [From expression of V_{bi}]

Similarly, minority carrier holes on the n-side:

$$p_{n0} = p_{p0} exp\left(-\frac{qV_{bi}}{kT}\right)$$

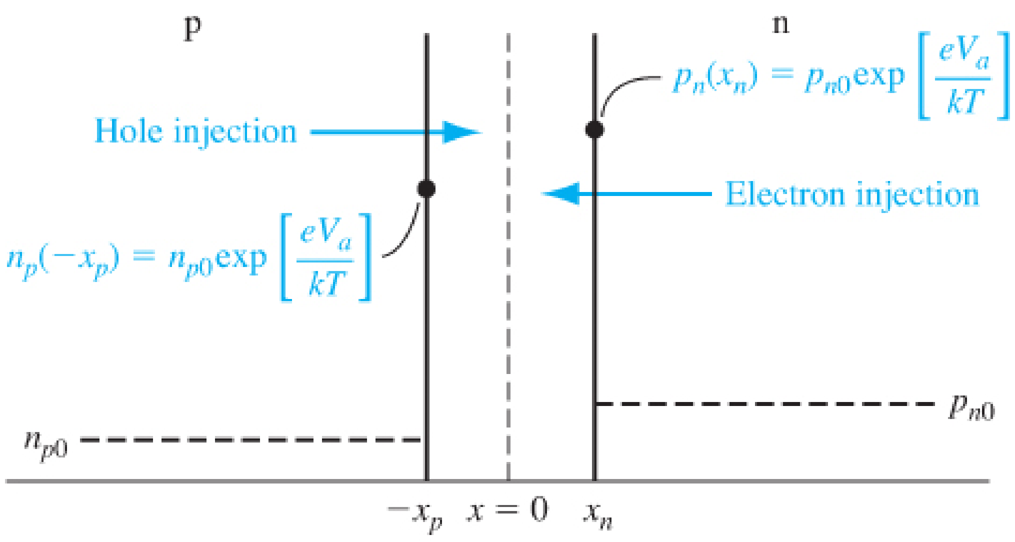

Figure 8.4 | Excess minority carrier concentrations at the space charge edges generated by the forward-bias voltage.

Ideal Characteristics: Minority Carrier Injection

- The application of voltage modulates the potential barrier:
  - V_{bi} becomes $(V_{bi} - V_a)$ [Assuming V_a is forward-biased]

- Diffusion of holes from p-side and electrons from n-side increases in the forward-bias

- Excess minority carriers get injected at the edge of depletion region

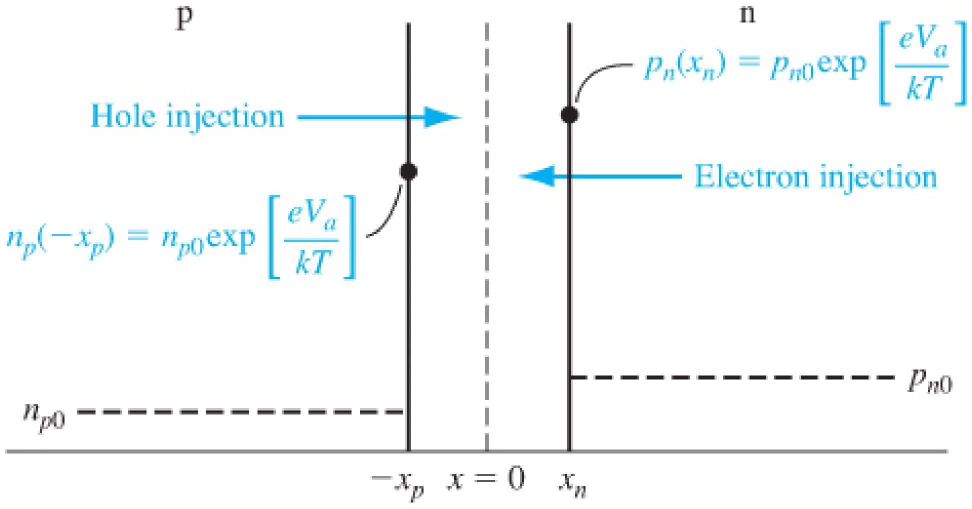

Figure 8.4 | Excess minority carrier concentrations at the space charge edges generated by the forward-bias voltage.

On p-side:

- At equilibrium: $n_{p0} = n_{n0} exp\left(-\frac{qV_{bi}}{kT}\right)$

- After applying bias: $n_p = n_{n0} exp\left(-\frac{q(V_{bi}-V_a)}{kT}\right)$

$$n_p = n_{n0} exp\left(-\frac{qV_{bi}}{kT}\right) exp\left(\frac{qV_a}{kT}\right) = n_{p0} exp\left(\frac{qV_a}{kT}\right)$$

Similarly on n-side:

- After applying bias: $p_n = p_{n0} exp\left(\frac{qV_a}{kT}\right)$

- Excess minority carriers are injected on the depletion region edges

- Then, they diffuse in the neutral region

PN Junction Minority Carrier Injection: Problem

Consider a PN junction at equilibrium at 300 K with $N_a = 6 \times 10^{15} cm^{-3}$ and $N_d = 10^{16} cm^{-3}$

Determine the minority carrier concentration at the edge of the space charge region, if a forward bias of 0.6 V is applied.

[Take $n_i = 1.5 \times 10^{10} cm^{-3}$]

$$n_{p0} = \frac{n_i^2}{p_{p0}} = 3.75 \times 10^4 cm^{-3}$$

$$p_{n0} = \frac{n_i^2}{n_{n0}} = 2.25 \times 10^4 cm^{-3}$$

$$n_p(-x_p) = n_{p0} exp\left(\frac{qV_a}{kT}\right) = 4.31 \times 10^{14} cm^{-3}$$

$$p_n(x_n) = p_{n0} exp\left(\frac{qV_a}{kT}\right) = 2.59 \times 10^{14} cm^{-3}$$

Ideal Characteristics: Diffusion of Excess Carriers

- The neutral region already has a large concentration of majority carriers
- The minority carrier and majority carrier *recombine* in the neutral region

- The concentration of minority carrier decays exponentially in the neutral region

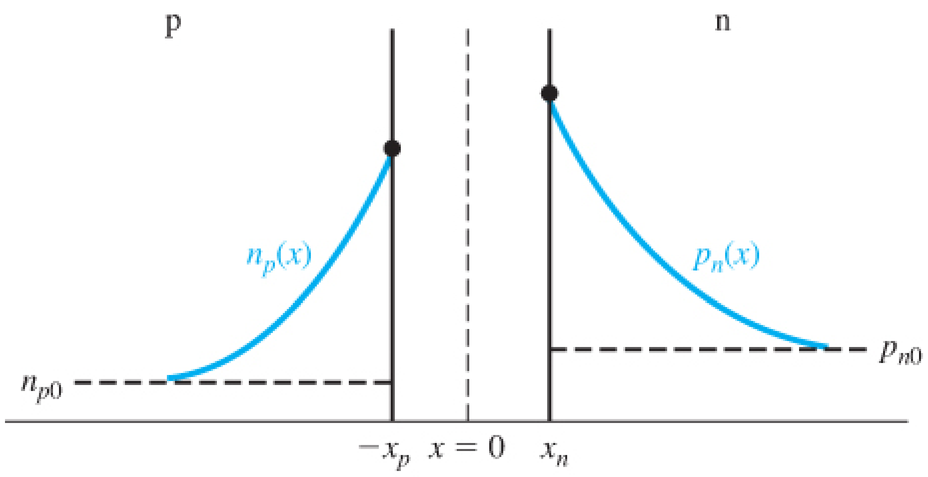

Figure 8.5 | Steady-state minority carrier concentrations in a pn junction under forward bias.

- The profile of the minority carriers can be found using ambipolar transport equation with the following boundary conditions:
  1. At the edges the concentrations are $n_{p0}exp\left(\frac{qV_a}{kT}\right)$ and $p_{n0}exp\left(\frac{qV_a}{kT}\right)$
  2. At far away from junction it is same as equilibrium minority concentration n_{p0} and p_{n0}

Ideal Characteristics: Hole and Electron Diffusion Currents

On n-side: $\delta p_n(x) = p_{n0}\left[exp\left(\frac{qV_a}{kT}\right) - 1\right] exp\left(\frac{x_n - x}{L_p}\right)$

On p-side: $\delta n_p(x) = n_{p0}\left[exp\left(\frac{qV_a}{kT}\right) - 1\right] exp\left(\frac{x + x_p}{L_n}\right)$

Problem: Compute the hole diffusion current in the n region at $x = x_n$. The hole diffusion current is given as:

$$J_p(x_n) = -qD_p \frac{dp_n(x)}{dx}\bigg|_{x=x_n} = -qD_p \frac{d\delta p_n(x)}{dx}\bigg|_{x=x_n}$$

Answer:

$$J_p(x_n) = \frac{qD_p p_{n0}}{L_p}\left[exp\left(\frac{qV_a}{kT}\right) - 1\right]$$

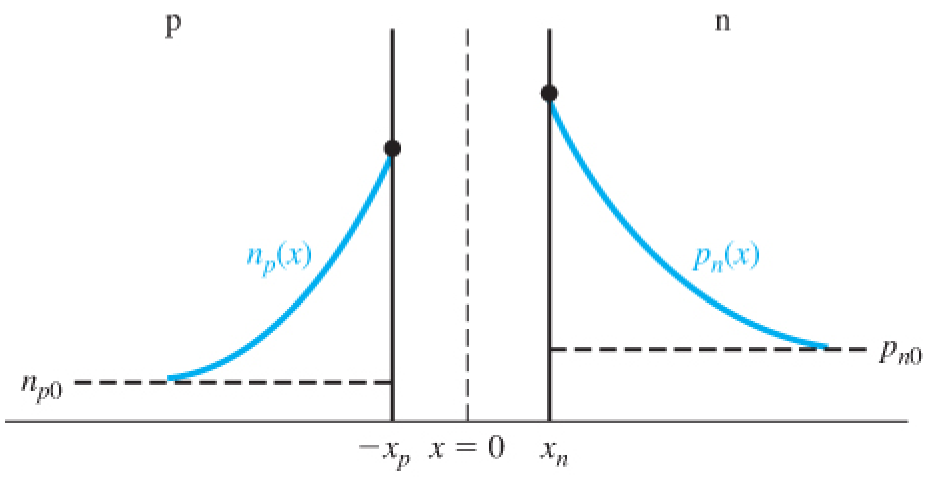

Figure 8.5 | Steady-state minority carrier concentrations in a pn junction under forward bias.

Similarly on the p-side:

$$J_n(-x_p) = \frac{qD_n n_{p0}}{L_n}\left[exp\left(\frac{qV_a}{kT}\right) - 1\right]$$

Ideal Characteristics: Total Diffusion Current

On n-side:

$$J_p(x_n) = \frac{qD_p p_{n0}}{L_p}\left[exp\left(\frac{qV_a}{kT}\right) - 1\right]$$

On the p-side:

$$J_n(-x_p) = \frac{qD_n n_{p0}}{L_n}\left[exp\left(\frac{qV_a}{kT}\right) - 1\right]$$

By the assumption of ideal PN junction, total current density:

$$J = J_p(x_n) + J_n(-x_p)$$

$$= \left[\frac{qD_p p_{n0}}{L_p} + \frac{qD_n n_{p0}}{L_n}\right]\left[exp\left(\frac{qV_a}{kT}\right) - 1\right]$$

Define reverse-saturation current density

$$J_s = \left[\frac{qD_p p_{n0}}{L_p} + \frac{qD_n n_{p0}}{L_n}\right]$$

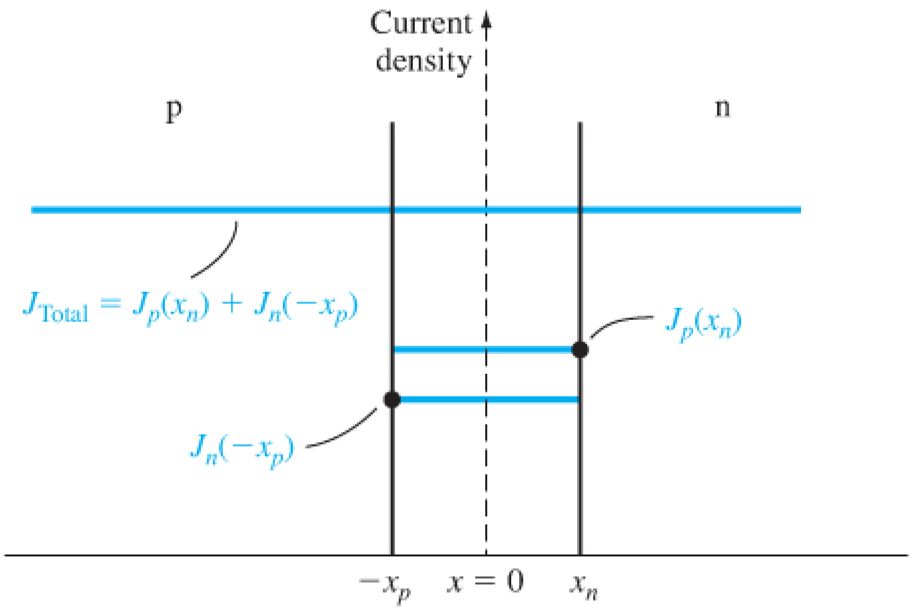

Figure 8.7 | Electron and hole current densities through the space charge region of a pn junction.

Ideal PN junction current:

$$J = J_s\left[exp\left(\frac{qV_a}{kT}\right) - 1\right]$$

Ideal Characteristics: Forward and Reverse Mode

Ideal PN junction current:

$$J = J_s\left[exp\left(\frac{qV_a}{kT}\right) - 1\right]$$

- Valid for both forward and reverse mode

- In forward mode, if V_a is greater than a few kT/q, we can ignore -1
  - ➢ The current increase exponentially

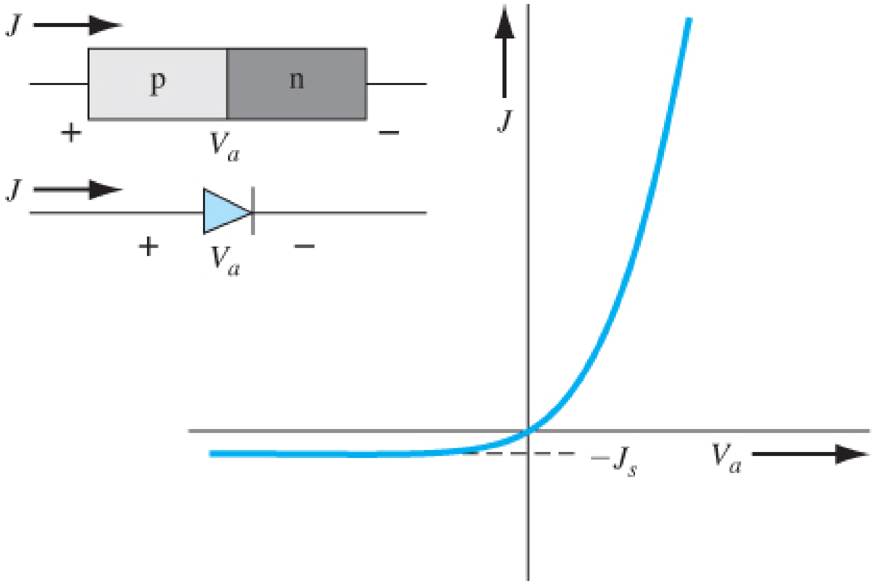

Figure 8.8 | Ideal I–V characteristic of a pn junction diode.

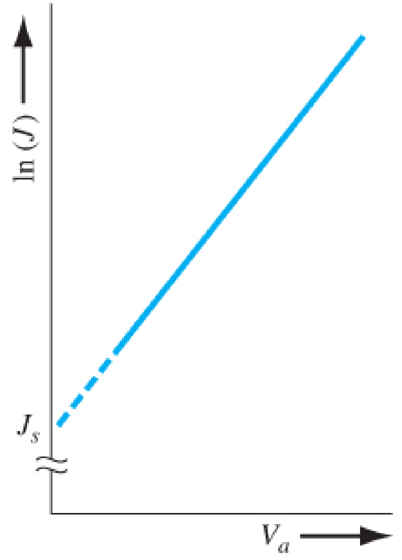

Figure 8.9 | Ideal I–V characteristic of a pn junction diode with the current plotted on a log scale.

- In reverse mode, if V_a is less than a few $-kT/q$, we can ignore the exponential term
  - ➢ The current saturates to J_s

Ideal Characteristics: Summary

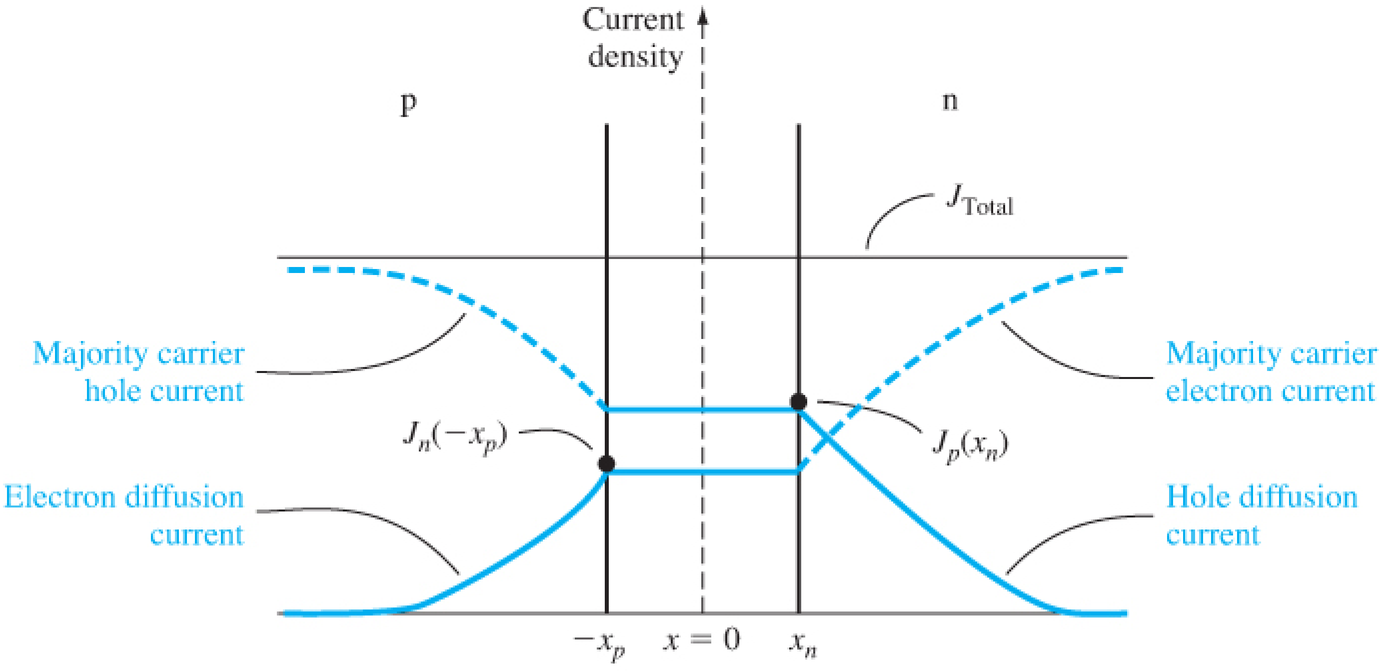

Figure 8.10 | Ideal electron and hole current components through a pn junction under forward bias.

- Minority carrier diffusion current densities decay with distance

- Total current density remains constant throughout the device

- Increase in majority carrier current densities compensate for the decreased diffusion current.

$$J_p(x_n) = \frac{qD_p p_{n0}}{L_p}\left[exp\left(\frac{qV_a}{kT}\right) - 1\right] exp\left(\frac{x_n - x}{L_n}\right)$$

Ideal Characteristics: Temperature Effect

Ideal PN junction current:

$$J = J_s\left[exp\left(\frac{qV_a}{kT}\right) - 1\right]$$

Reverse-saturation current density

$$J_s = \left[\frac{qD_p p_{n0}}{L_p} + \frac{qD_n n_{p0}}{L_n}\right]$$

- Reverse saturation current depends on minority carriers
  - It is dependent on n_i^2
  - n_i^2 is a strong function of temperature
  - J_s depends on temperature

- For silicon, ideal reverse saturation current density increase $4\times$ for every $10℃$ increase in temperature

- Forward-bias current also increase due to increase in temperature
  - Due to increase in J_s
  - Less sensitive to temperature because of $exp\left(\frac{qV_a}{kT}\right)$ term

PN Junction

Non-Ideal Current-Voltage Characteristics

I-V Characteristics: Non-idealities

Ideal PN junction current:

$$J = J_s\left[exp\left(\frac{qV_a}{kT}\right) - 1\right]$$

- Generation-recombination current
- High-level injection

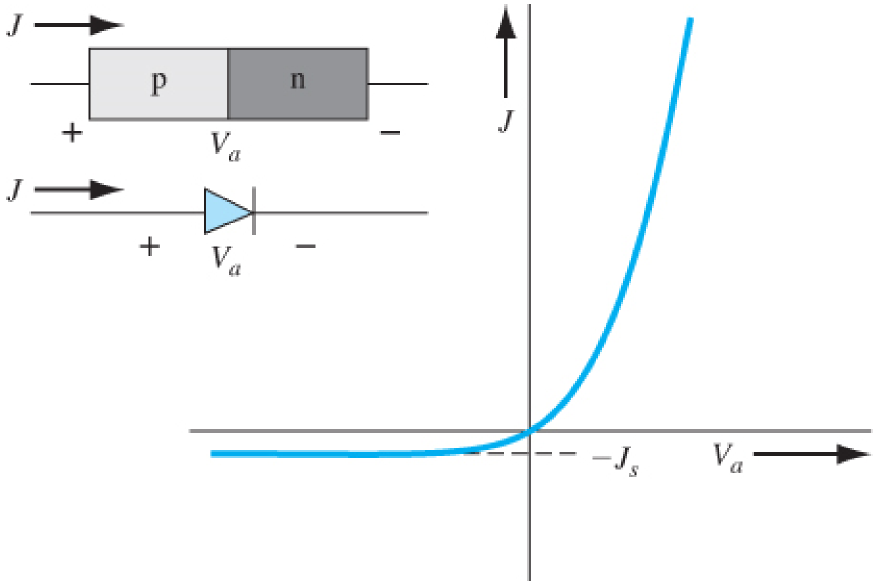

Figure 8.8 | Ideal *I–V* characteristic of a pn junction diode.

Generation/Recombination Current

- The equilibrium is disturbed in the depletion region $np \neq n_i^2$
- Generation/recombination of carriers in the depletion region tries to restore the equilibrium
- These effects increase the current of a diode from the ideal PN junction current

Reverse-biased diode:

- Carriers move out of the depletion region
- Less carriers in the depletion region
- Generation of carriers in the depletion region tries to restore the equilibrium

Forward-biased diode:

- Minority carriers injected through the depletion region
- Excess carriers in the depletion region
- Recombination of carriers in the depletion region tries to restore the equilibrium

Generation Current (Reverse-biased)

- SRH recombination rate:

$$R = \frac{(np - n_i^2)}{\tau_{po}(n + n') + \tau_{n0}(p + p')}$$

- In reverse-biased mode: the carrier concentration in the depletion region is $n \approx p \approx 0$

- SRH recombination rate

$$R = \frac{-n_i^2}{\tau_{po}n' + \tau_{n0}p'}$$

- Negative recombination means generation

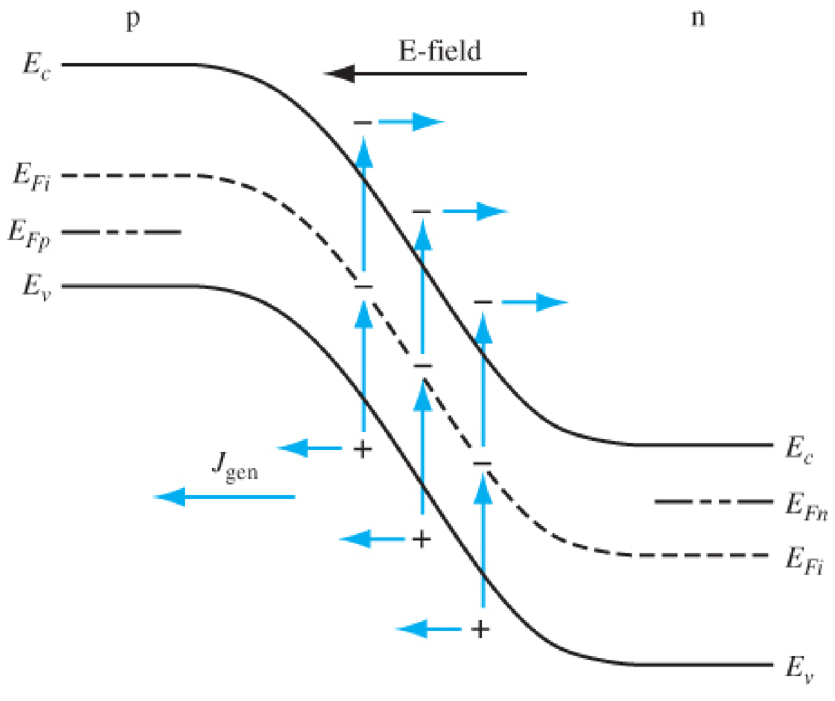

Figure 8.12 | Generation process in a reverse-biased pn junction.

- Electric field sweeps out generated carriers in the direction of the reverse diode current

- These generated carriers produce extra current in the reverse-biased mode (in addition to the ideal reverse saturation current)

Generation Current Calculation (Reverse-biased)

- Total reverse-biased current density is:

$$J_R = J_s + J_{gen}$$

- J_{gen} is a function of reverse bias (because W is a function of reverse bias and increases in W increases J_{gen})
  - ➢ reverse-biased current density increases with V_R

- For silicon, J_{gen} can be 4 orders of magnitude higher than ideal revserse saturation current density J_s
  - ➢ Generation current is dominant reverse-biased current

Recombination Rate (Forward-biased)

- Both excess electrons and holes are present in the depletion region
- Some electron-hole pairs recombine
- SRH recombination rate:

$$R = \frac{\left(np - n_i^2\right)}{\tau_{po}(n + n') + \tau_{n0}(p + p')}$$

We can derive the following :

$J_{rec} = J_{r0} exp\left(\frac{qV_a}{2kT}\right)$ where $J_{r0} = \frac{qWn_i}{2\tau_o}$

Total Forward Bias Current

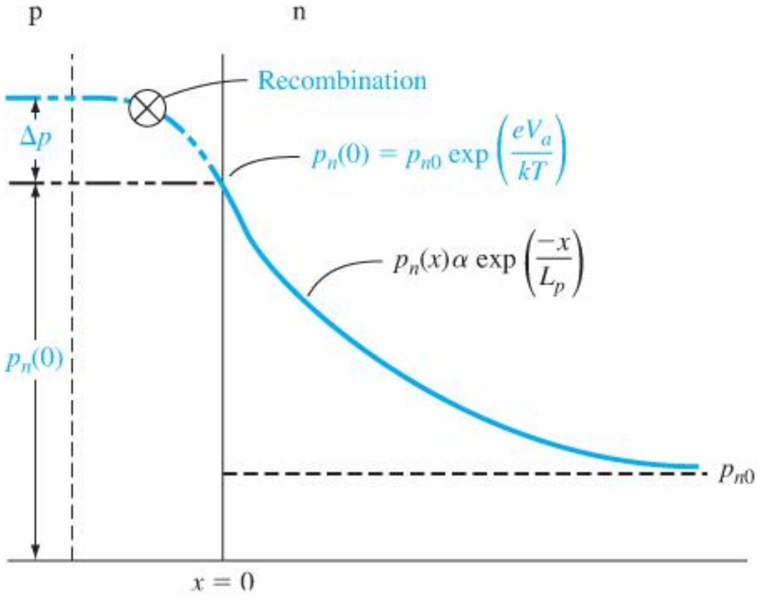

Figure 8.15 | Because of recombination, additional holes from the p region must be injected into the space charge region to establish the minority carrier hole concentration in the n region.

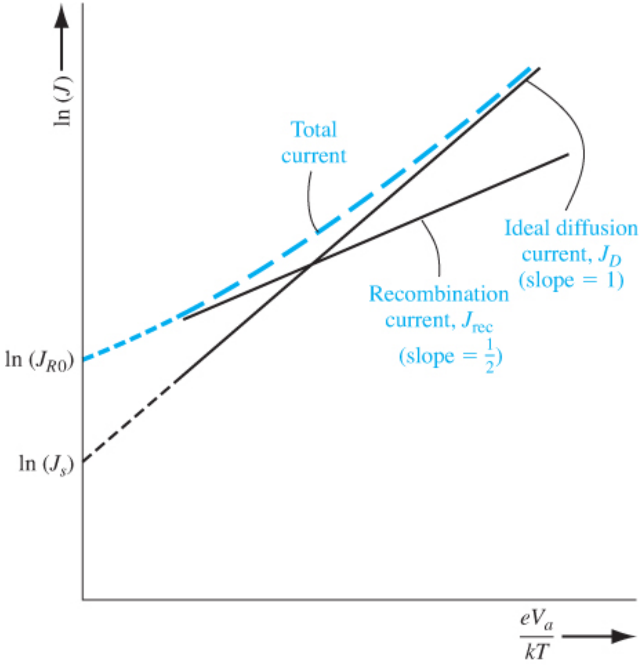

Figure 8.16 | Ideal diffusion, recombination, and total current in a forward-biased pn junction.

Combine them, we can write current equation with a fitting parameter η

$$I = I_s\left[exp\left(\frac{qV_a}{\eta kT}\right) - 1\right]$$

- $1 < \eta < 2$
- $\eta = 1$ for diffusion-dominant
- $\eta = 2$ for recombination-dominant

$$J = J_{rec} + J_D = J = J_{r0}exp\left(\frac{qV_a}{2kT}\right) + J_s exp\left(\frac{qV_a}{kT}\right)$$

Low bias: recombination current dominates, High bias: diffusion current dominates

High-level Injection

- If forward bias is increased, the concentration of minority carrier injected will be more than equilibrium majority carriers

- Consider the charge concentrations after excess carriers are injected,
- By definition $n = n_0 + \delta n$ and $p = p_0 + \delta p$,
- Also, following conditions hold (on the n-side)
  - ➢ $\delta p \gg n_0$ (because of high-level injection)
  - ➢ $n_0 \gg p_0$ (because it is extrinsic)
  - ➢ Also $\delta n = \delta p$ (due to charge neutrality)

$$np = (n_0 + \delta n)(p_0 + \delta p)$$
$$np \approx \delta p^2$$

- We know that: $np = n_i^2 exp\left(\frac{qV_a}{kT}\right)$
- So, $\delta p = \delta n = n_i exp\left(\frac{qV_a}{2kT}\right)$

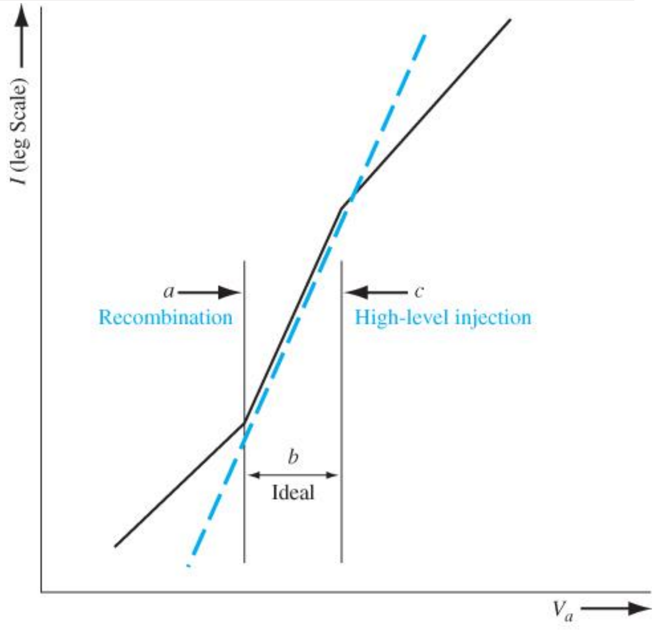

Figure 8.17 | Forward-bias current versus voltage from low forward bias to high forward bias.

Current rises slowly in high-level injection

$$I \propto exp\left(\frac{qV_a}{2kT}\right)$$

PN Junction

Diffusion Capacitance and Resistance

Diffusion Capacitance: Physical Interpretation

Assume that we superimpose a small AC signal over a DC signal for a diode in forward biased mode:

$$V_a = V_{dc} + \hat{v}sin\omega t$$

How will the hole concentration change at the edge of the depletion region on the n-side?

At $t = t_0$, $V_a = V_{dc}$

$$p_n(t_0) = p_{no}exp\left[\frac{V_{dc}}{V_t}\right]$$

Increases in +ve half cycle and reaches peak at $t = t_1$

Decreases in -ve half cycle and reaches lowest value at $t = t_2$

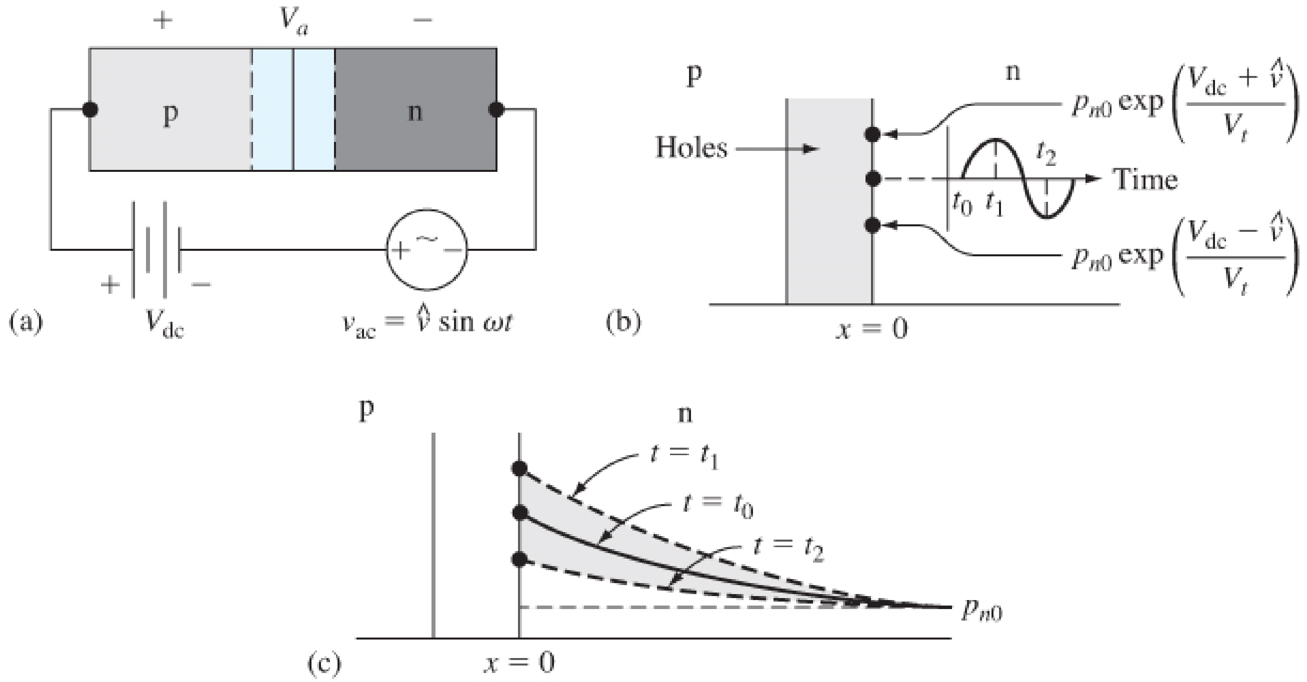

Figure 8.19 | (a) A pn junction with an ac voltage superimposed on a forward-biased dc value; (b) the hole concentration versus time at the space charge edge; (c) the hole concentration versus distance in the n region at three different times.

Diffusion Capacitance: Definition

- The injected holes diffuse in the n region and get recombined with the majority carrier electrons
- Assuming that the frequency of AC signal is not high compared to the time required to diffuse, a steady-state hole concentration can be reached

- The charge ΔQ is being charged/discharged through the junction

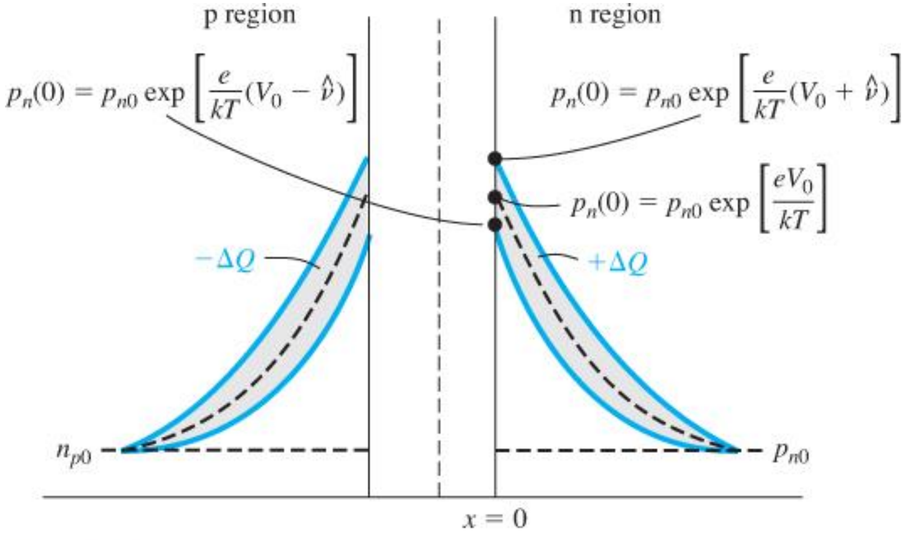

Figure 8.21 | Minority carrier concentration changes with changing forward-bias voltage.

- We can define ***diffusion capacitance*** as:

$$C_d = \left.\frac{dQ}{dV}\right]_{V_{dc}}$$

[similar to junction capacitance]

Differences of diffusion cap. with depletion cap.:

- Physical origins are different (as demonstrated)
- It is 3-4 order magnitude larger
- It is important in forward-biased mode only

References

- D. A. Neaman, *"Semiconductor Physics and Devices"*